

K-FIELD

3 x 2 x 2 SOLAR-SYSTEM BODY ROTATION-PERIOD HARMONIC LADDER

The Saturn - Neptune - Earth 3/2 temporal sequence and its reciprocal 2/3 frequency ladder

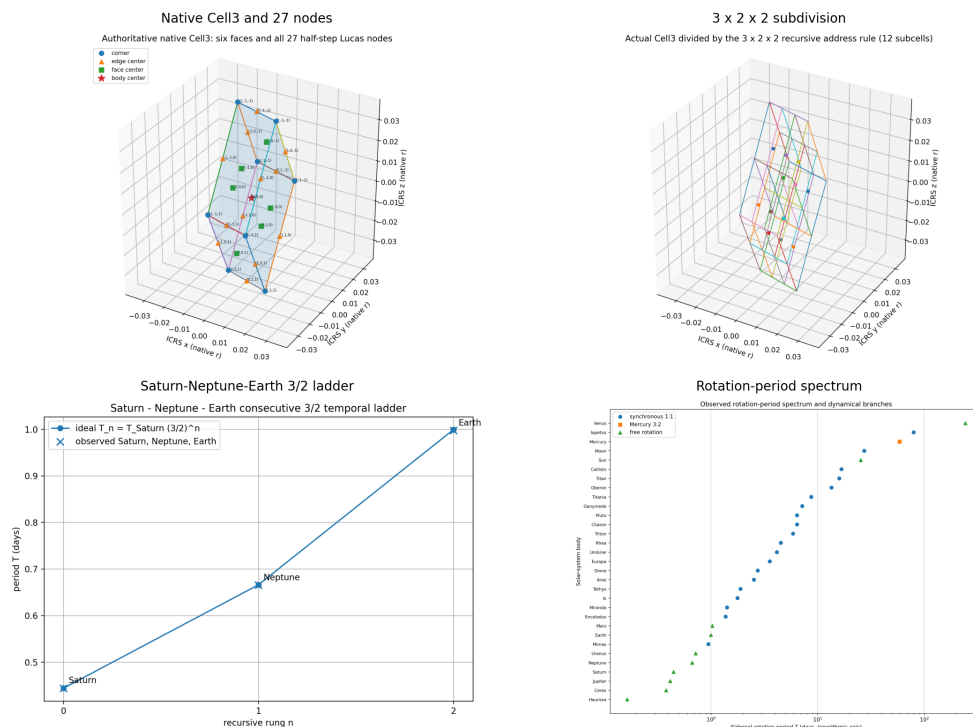


Figure 1. Title composite. Every panel is generated from the authoritative Cell3 packet or the actual solar-system rotation-period ledger reproduced in this report.

Principal Investigator

P. Dwayne Esterline
BS, Huntington University (1999)
MBA, Indiana Wesleyan University (2008)

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Frontispiece: authoritative base cell, face planes, and all nodes

Authoritative native Cell3: six faces and all 27 half-step Lucas nodes

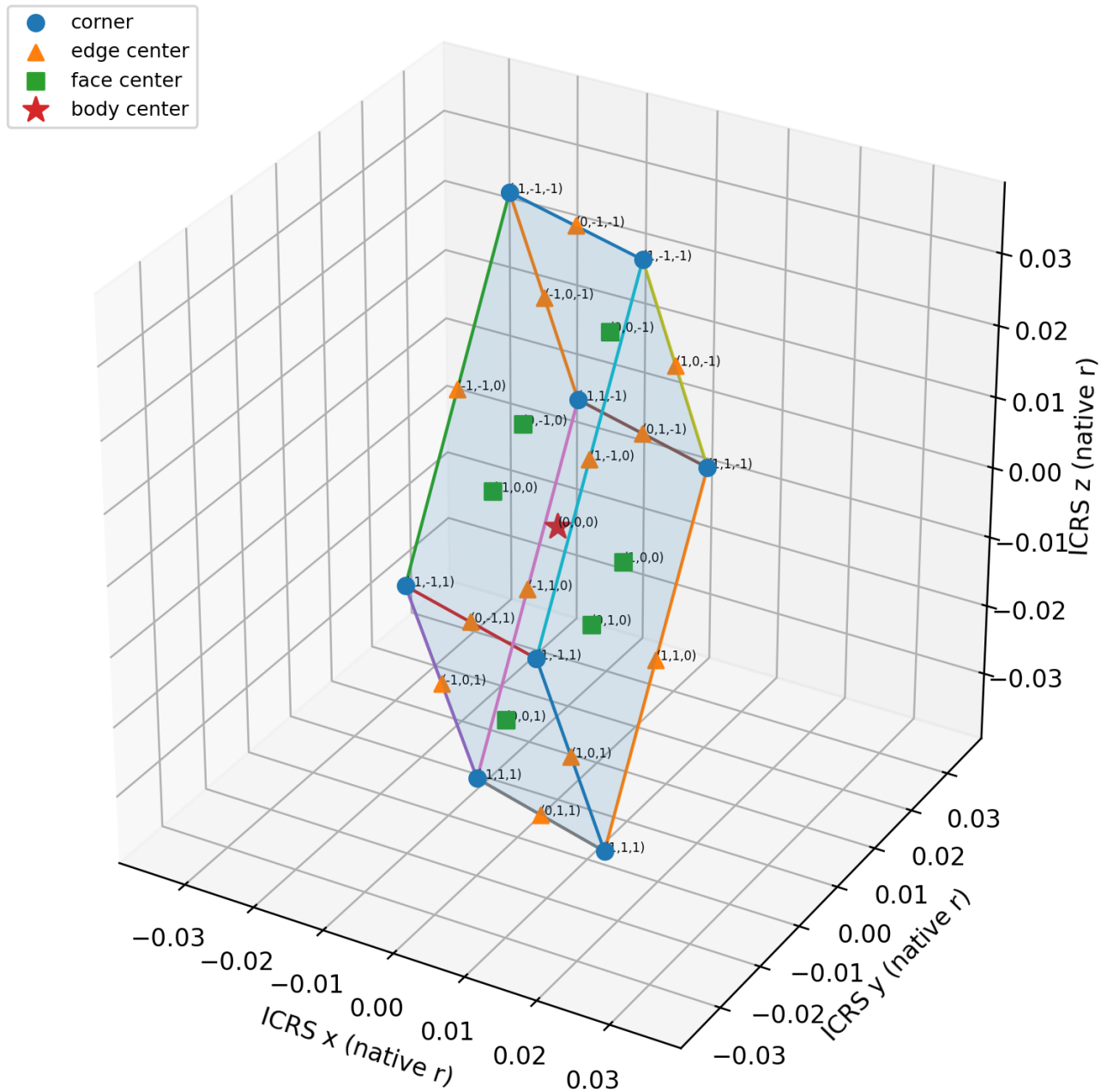


Figure 2. Native ICRS Cell3 and all 27 half-step Lucas nodes. The eight corners, twelve edge centers, six face centers, and body center are located by $p(i,j,k)=B[i,j,k]^T/2$ with i,j,k in $\{-1,0,1\}$. The axes share one native scale; the geometry is not schematic.

Six boundary face planes

Six dimensionally correct Cell3 boundary face planes and their 3 x 3 node sections

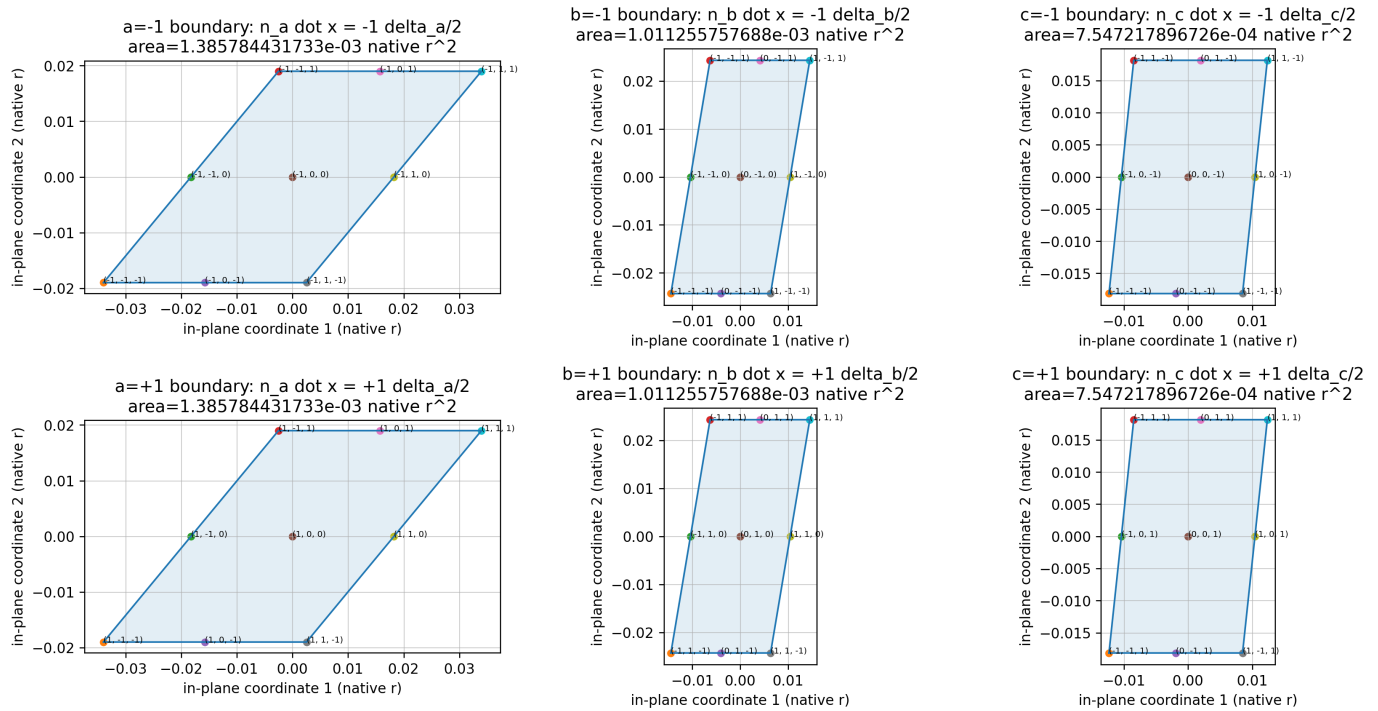


Figure 3. The six centered Cell3 boundaries $n_a \cdot x = \pm \Delta_a/2$, $n_b \cdot x = \pm \Delta_b/2$, and $n_c \cdot x = \pm \Delta_c/2$. Each face contains the actual 3 x 3 half-step node section generated by the same basis.

The 3 x 2 x 2 recursive subdivision

Actual Cell3 divided by the 3 x 2 x 2 recursive address rule (12 subcells)

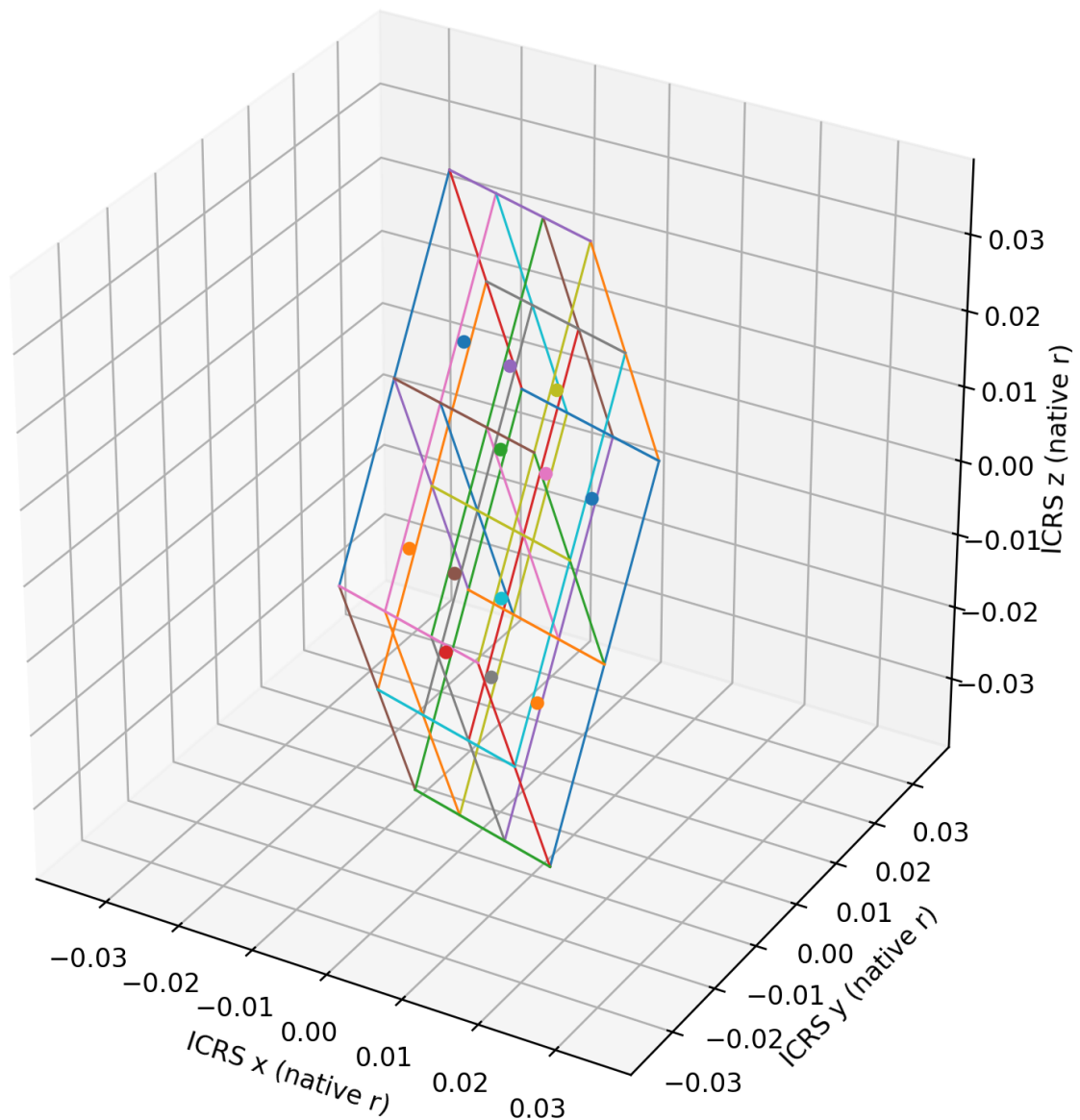


Figure 4. The authoritative Cell3 divided into three equal address intervals along the a coordinate and two equal intervals along each of b and c. The mapped wireframe contains twelve dimensionally correct subcells.

The direct interval quotient is $(1/2)/(1/3)=3/2$. Its reciprocal quotient is $(1/3)/(1/2)=2/3$. These two numbers are the direct and reciprocal temporal scale factors developed in this report.

Abstract

The solar-system body rotation-period ledger separates into three directly readable dynamical branches. Twenty bodies occupy synchronous or mutually synchronous 1:1 states. Mercury occupies the 3:2 spin-orbit state. The remaining free-rotation branch contains a consecutive Saturn - Neptune - Earth ladder whose period ratios are 1.498307718 and 1.499061848, placing Neptune and Earth on the first and second $3/2$ rungs above Saturn. The Earth-to-Saturn ratio is 2.246055936, close to $9/4=(3/2)^2$. In reciprocal frequency coordinates the same sequence descends by $2/3$ per rung. The ratio $3/2$ is generated directly by the Cell3 $3 \times 2 \times 2$ recursive subdivision: it is the quotient of the binary and ternary interval scales. This establishes a temporal reading of the recursive geometry in which rotation periods occupy direct scale states and rotation frequencies occupy their reciprocal states. The report reproduces the authoritative Cell3 basis, six plane boundaries, all 27 half-step nodes, the full 31-body period ledger, resonance relations, Cell3 harmonic/MOD assignments, and a machine-readable archival packet.

Canonical determination reproduced in full

Correct. The connection is structural:

$$T_{(n+1)} = (3/2) T_n$$

The factor $3/2$ is the quotient of the ternary and binary Cell3 recursive subdivision factors. The Saturn-Neptune-Earth sequence therefore occupies consecutive recursive rungs:

$$T_{\text{Saturn}} = T_0, \quad T_{\text{Neptune}} \text{ approximately } T_0(3/2), \quad T_{\text{Earth}} \text{ approximately } T_0(3/2)^2.$$

Equivalently, the angular frequencies descend by the reciprocal factor:

$$\omega_{(n+1)} = (2/3) \omega_n.$$

The Earth-Saturn ratio consequently becomes

$$T_{\text{Earth}}/T_{\text{Saturn}} \text{ approximately } 9/4 = (3/2)^2.$$

This is the temporal form of the Cell3 $3 \times 2 \times 2$ recursive grammar, not merely a generic harmonic relation. The recursive construction explicitly contains ternary and binary refinement factors, while the reciprocal shell grammar organizes states through a 1:2:3 ladder.

The correct interpretation is therefore: solar-system free-rotation periods can occupy discrete temporal scale states generated by repeated $3/2$ Cell3 recursion, while synchronous satellites form a separate orbitally constrained 1:1 branch. The remaining free rotators should be tested as displaced or alternate-family recursive addresses, with their residual phase compared directly against spin-axis position in the K-Field.

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1. Coordinate, unit, and data contract

1.1 Native-before-rendered rule

Cell3 is defined first in native unit-sphere-derived coordinates. The formal radius gauge r carries native length. Basis vectors therefore have dimension r , face areas have dimension r^2 , and volume has dimension r^3 . Angles, ratios, lattice addresses, and normalized reciprocal radial amplitudes are dimensionless. Decimal values are terminal witnesses of the native algebra and do not feed back into the definition.

$$\begin{aligned} B_r &= r \, N^{(-1)} \, D_{\text{delta}} \\ N \, B_r &= r \, D_{\text{delta}} \\ B_V &= B_r / |\det(B_r)|^{(1/3)} \end{aligned}$$

The ordered embedding is left-handed: $\det(B) < 0$. The basis order a, b, c and the ICRS x, y, z order are part of the coordinate contract.

1.2 Rotation-period convention

For each body represented by a linear prime-meridian coefficient W_{dot} in the NAIF/IAU orientation model, the period used here is the magnitude of one full modeled rotation:

$$T = 360 \, \text{deg} / |W_{\text{dot}}|.$$

This definition preserves retrograde direction in the sign of W_{dot} while using a positive period magnitude. The Sun and giant planets are represented by their adopted prime-meridian systems; they are not treated as perfectly rigid surfaces. Haumea is entered from the JPL physical-parameter period of 0.1631 day.

1.3 Deterministic analysis rule

The work consists of direct transformations: period ratios, exact rational targets, recursive rung addresses, reciprocal frequencies, and signed residuals. The conclusions are read from those quantities without randomization, p -values, or proof-statistical substitution.

2. Authoritative Cell3 geometry

2.1 Native ICRS basis

The current-master native basis columns expressed in the orthonormal ICRS frame are:

$$B = \begin{bmatrix} +0.02038143851694645, & +0.00219874596832336, & -0.00063037948534737, \\ -0.00341641449641055, & +0.01290217025869254, & -0.02425772201712787, \\ -0.00233296591672896, & -0.03406650153605571, & -0.04292435457109382 \end{bmatrix}$$

Quantity	Value	Native dimension
a	0.020797058782433	r
b	0.036494205130727	r
c	0.049308565899859	r
angle(a,b)	83.933483012313 deg	angle
angle(a,c)	80.448130018338 deg	angle
angle(b,c)	50.363231813756 deg	angle
det(B)	-2.842066937809648567e-05	r ³
A_ab	7.547217896726240299e-04	r ²
A_ac	1.011255757688065251e-03	r ²
A_bc	1.385784431733287072e-03	r ²

2.2 Plane-normal duality

The three labelled unit normals and native spacings define six centered boundary planes. The duality $N B = D_{\text{delta}}$ means that moving one full basis vector changes exactly one plane-family coordinate by its corresponding spacing.

$$\begin{aligned} n_a \cdot x &= \pm \text{delta}_a/2 \\ n_b \cdot x &= \pm \text{delta}_b/2 \\ n_c \cdot x &= \pm \text{delta}_c/2 \end{aligned}$$

Family	ICRS unit normal (x,y,z)	RA, Dec (deg)	delta at r=1	Centered boundaries
a	(+0.995965190309, -0.083602162507, +0.032619290564)	(355.201792089, 1.869279271)	0.020508723238108	$n_a \cdot x = \pm 0.010254361619054$
b	(+0.089052594102, +0.866576769291, -0.491033948326)	(84.132672287, -29.408562676)	0.028104333806783	$n_b \cdot x = \pm 0.014052166903391$
c	(-0.194092200808, -0.913177168821, -0.358379234791)	(258.000576203, -21.000692601)	0.037657147000386	$n_c \cdot x = \pm 0.018828573500193$

2.3 Twenty-seven half-step nodes

The centered Lucas node set is generated by all addresses (i,j,k) in $\{-1,0,1\}^3$. The categories are fixed by the number of nonzero address components: three gives a corner, two an edge center, one a face center, and zero the body center.

$$p(i,j,k) = (1/2) B [i,j,k]^T.$$

3. The 3 x 2 x 2 recursive grammar

3.1 Address subdivision

One Cell3 address interval is divided into three parts along one labelled coordinate and two parts along each of the other two coordinates. The resulting partition contains $3 \times 2 \times 2 = 12$ subcells. Because the physical embedding is linear, the address subdivision maps into twelve geometrically similar subregions of the skewed Cell3 without altering incidence, orientation, or plane-family identity.

```
ternary interval = 1/3
binary interval = 1/2
subcell count = 3 x 2 x 2 = 12
```

3.2 Direct 3/2 ratio

Comparing one binary interval with one ternary interval gives the direct scale ratio:

$$(1/2) / (1/3) = 3/2.$$

Repeated scale transfer produces the ladder $T_n = T_0(3/2)^n$. One positive rung increases the direct period scale by 50 percent. One negative rung reduces it by one third.

3.3 Reciprocal 2/3 ratio

Frequency is reciprocal to period. The same recursive transition therefore reads as 2/3 in frequency space:

```
f = 1/T
T_(n+1) = (3/2) T_n
f_(n+1) = (2/3) f_n
omega_(n+1) = (2/3) omega_n.
```

The 3/2 period ladder and 2/3 frequency ladder are not two unrelated patterns. They are dual readings of one recursive state transition.

4. Rotation period and angular frequency

4.1 Period as a direct temporal scale

A rotation period T measures elapsed time per completed cycle. A larger T is a slower rotator. When periods occupy the recursive relation $T_n = T_0(3/2)^n$, each rung represents the same multiplicative change in temporal scale regardless of the chosen time unit.

4.2 Frequency as the reciprocal coordinate

Ordinary frequency f counts cycles per unit time, while angular frequency ω counts radians per unit time:

$$f = 1/T$$

$$\omega = 2\pi f = 2\pi/T.$$

Under a $3/2$ period increase, both f and ω decrease by $2/3$. This direct-reciprocal pairing matches the role of direct and reciprocal geometry in Cell3: a scale enlargement in one description is a scale contraction in the dual description.

4.3 Logarithmic rung coordinate

A multiplicative ladder becomes an additive integer address after taking a logarithm base $3/2$:

$$n(T) = \log(T/T_0) / \log(3/2).$$

Exact ladder states have integer n . The signed fractional part $n - \text{round}(n)$ is the logarithmic phase displacement from the nearest recursive rung.

5. Solar-system rotation-period dataset

5.1 Body set and sources

The body set contains the 31 objects for which the preceding solar-system spin-axis ledger supplied usable pole solutions. Eris and Makemake remain outside the computation because the source axis table did not provide adopted spin-pole solutions. Thirty periods are derived from NAIF pck00011.tpc prime-meridian rates. Haumea is entered from the JPL physical-parameter table.

5.2 Period spectrum

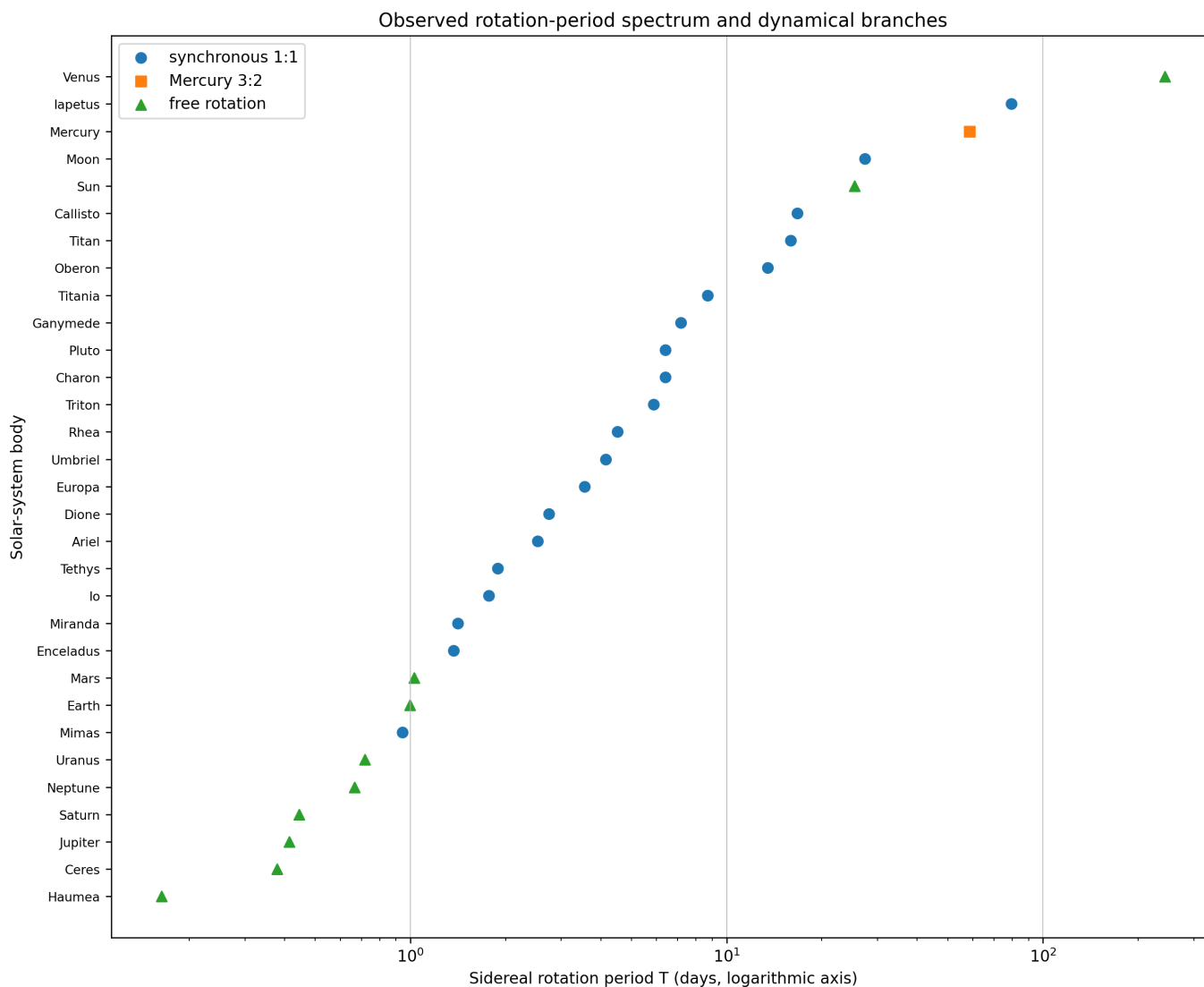


Figure 5. The 31-body period spectrum. Marker shape separates synchronous 1:1 states, Mercury 3:2, and the free-rotation branch. The logarithmic horizontal scale preserves the full range from Haumea at 0.1631 day to Venus at 243.018484 days.

5.3 Branch inventory

Branch	Body count	Defining relation
Synchronous or mutually synchronous 1:1	20	rotation period equals orbital period
Mercury 3:2 spin-orbit	1	three rotations per two solar orbits
Free rotation	10	period not assigned by 1:1 satellite locking in this ledger

6. Dynamical quantization branches

6.1 Synchronous 1:1 branch

Twenty bodies occupy the 1:1 state. In a synchronous state, one axial rotation is completed during one orbital revolution, so the same hemisphere remains directed toward the primary. The branch includes the Moon; the four Galilean satellites in the supplied set; the major Saturnian and Uranian satellites listed in the period ledger; Triton; and the mutually locked Pluto-Charon pair.

$$T_{\text{rotation}} / T_{\text{orbit}} = 1.$$

This branch is discrete, but it is orbitally constrained. It must therefore be kept separate from the free-rotation ladder when reading a field-wide temporal scale sequence.

6.2 Mercury 3:2 branch

Mercury completes three rotations for every two revolutions about the Sun. Using the adopted rotation period 58.646146 days and the orbital period used in the computational packet gives a spin-frequency to orbital-frequency ratio of 1.500000653.

$$f_{\text{spin}} / f_{\text{orbit}} = 3/2.$$

Mercury demonstrates that the 3/2 state already functions as a stable solar-system rotational commensurability. It is dynamically bound to orbital motion and is therefore distinguished from the Saturn-Neptune-Earth free-rotation sequence.

6.3 Galilean 1:2:4 chain

Io, Europa, and Ganymede are synchronously rotating satellites whose orbital periods occupy the Laplace sequence. Their rotation periods therefore inherit the same approximate 1:2:4 ladder. The measured period ratios in the adopted model are Europa/Io=2.007294785, Ganymede/Europa=2.014696790, and Ganymede/Io=4.044090359.

6.4 Saturnian 2:1 pairs

The adopted periods give Tethys/Mimas=2.003139720 and Dione/Enceladus=1.997430617. These pairs show the binary branch directly. The Enceladus-Dione relation is also the orbital resonance that maintains Enceladus tidal activity.

6.5 Pluto-Charon mutual 1:1 state

Pluto and Charon both have the adopted period 6.387223 days in the ledger. Their ratio is exactly 1.000000000 at the precision carried here. The same surfaces face one another continuously.

7. Direct resonance ledger

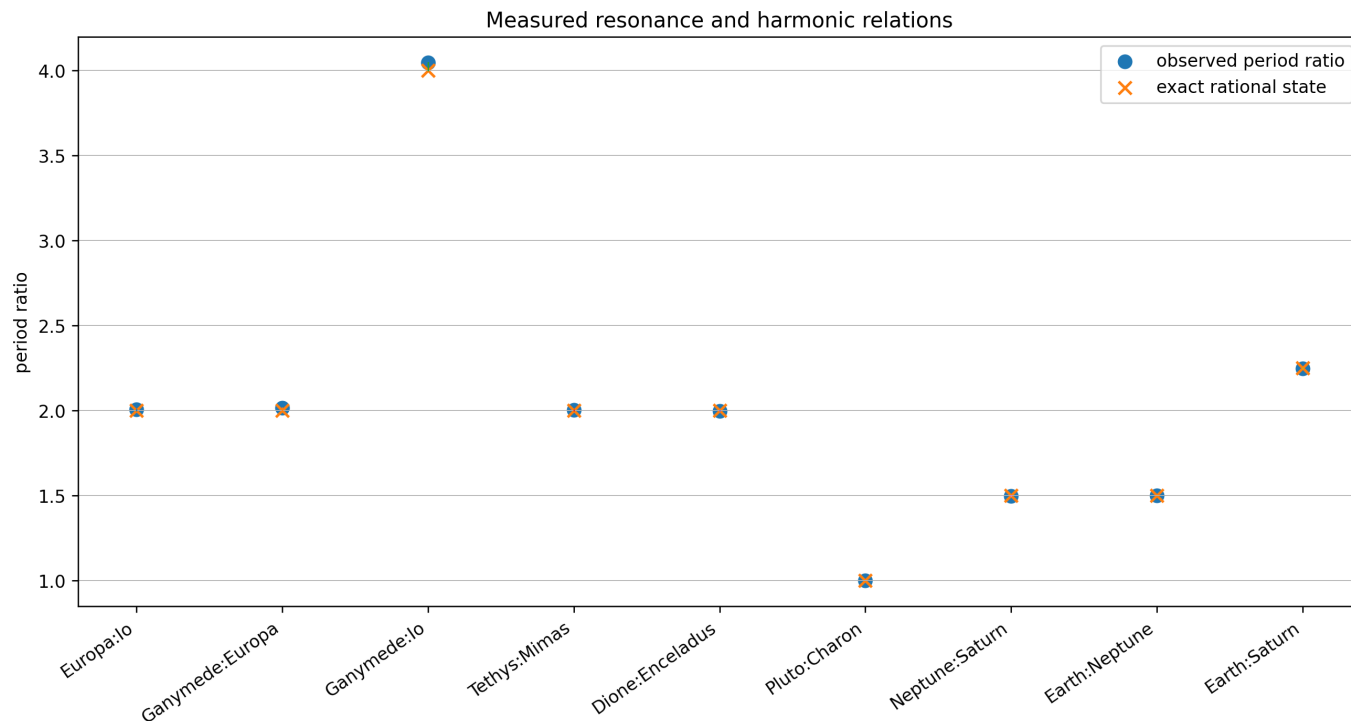


Figure 6. Observed period ratios compared with their exact rational states. Vertical segments are direct signed displacements from the target ratio.

Relation	Observed ratio	Target	Residual (%)	MOD-81 q	rho
Europa:Io	2.007294785	2.000000000	+0.364739	1	-0.005050894
Ganymede:Europa	2.014696790	2.000000000	+0.734839	1	+0.002351111
Ganymede:Io	4.044090359	4.000000000	+1.102259	4	-0.005292357
Tethys:Mimas	2.003139720	2.000000000	+0.156986	0	+0.003139720
Dione:Enceladus	1.997430617	2.000000000	-0.128469	0	-0.002569383
Pluto:Charon	1.000000000	1.000000000	+0.000000	0	+0.000000000
Neptune:Saturn	1.498307718	1.500000000	-0.112819	0	-0.001692282
Earth:Neptune	1.499061848	1.500000000	-0.062543	0	-0.000938152
Earth:Saturn	2.246055936	2.250000000	-0.175292	0	-0.003944064

8. The Saturn - Neptune - Earth 3/2 ladder

8.1 Consecutive direct period states

Saturn is assigned rung $n=0$. One $3/2$ transition predicts the Neptune rung; a second predicts the Earth rung.

$$T_0 = T_{\text{Saturn}}$$

$$T_1 = (3/2) T_0$$

$$T_2 = (3/2)^2 T_0 = (9/4) T_0.$$

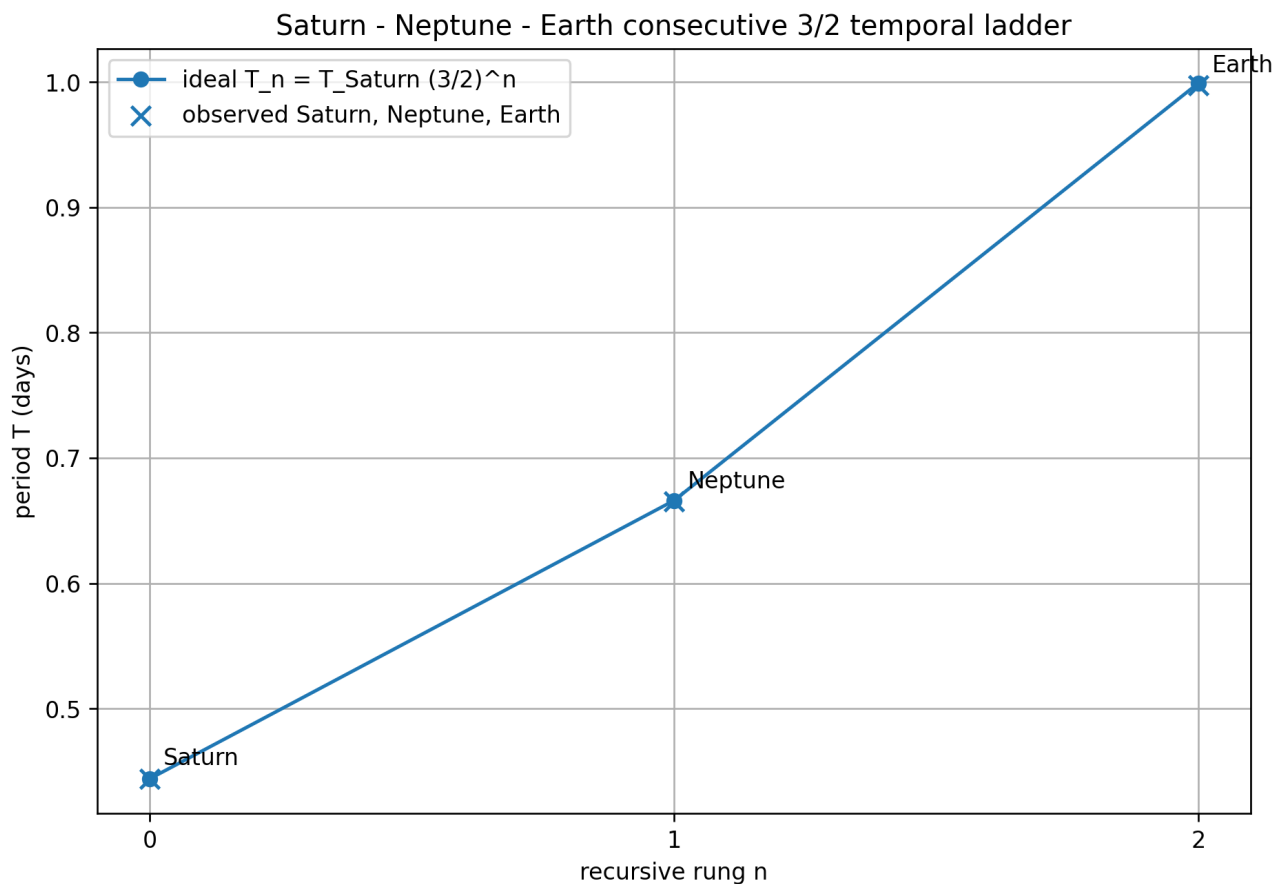


Figure 7. The actual Saturn, Neptune, and Earth periods plotted against the ideal consecutive $3/2$ ladder generated from Saturn.

Body	n	Actual T (day)	Ideal T (day)	Difference (min)	Residual (%)	log ₂ (3/2) address
Saturn	0	0.444009259239	0.444009259239	+0.000000	+0.0000000000	0.000000000000
Neptune	1	0.665262499942	0.666013888858	-1.082000	-0.112818806	0.997215975136
Earth	2	0.997269632263	0.999020833287	-2.521729	-0.175291742	1.995672980127

8.2 Measured ratios

Ratio	Observed	Recursive value	Signed displacement	Relative displacement
$T_{\text{Neptune}}/T_{\text{Saturn}}$	1.498307717911	1.500000000000	-0.001692282089	-0.112818806%
$T_{\text{Earth}}/T_{\text{Neptune}}$	1.499061847542	1.500000000000	-0.000938152458	-0.062543497%
$T_{\text{Earth}}/T_{\text{Saturn}}$	2.246055935798	2.250000000000	-0.003944064202	-0.175291742%

The first step differs from $3/2$ by -0.112819 percent. The second differs by -0.062543 percent. Across two steps, Earth differs from the $9/4$ Saturn state by -0.175292 percent. The sequence is therefore resolved as two consecutive near-integer recursive rung transitions rather than one isolated pairwise coincidence.

9. Reciprocal 2/3 frequency ladder

9.1 Frequency transformation

The direct period ladder reverses in reciprocal frequency coordinates. Saturn has the highest frequency of the three; Neptune is near 2/3 of Saturn; Earth is near another factor of 2/3 below Neptune.

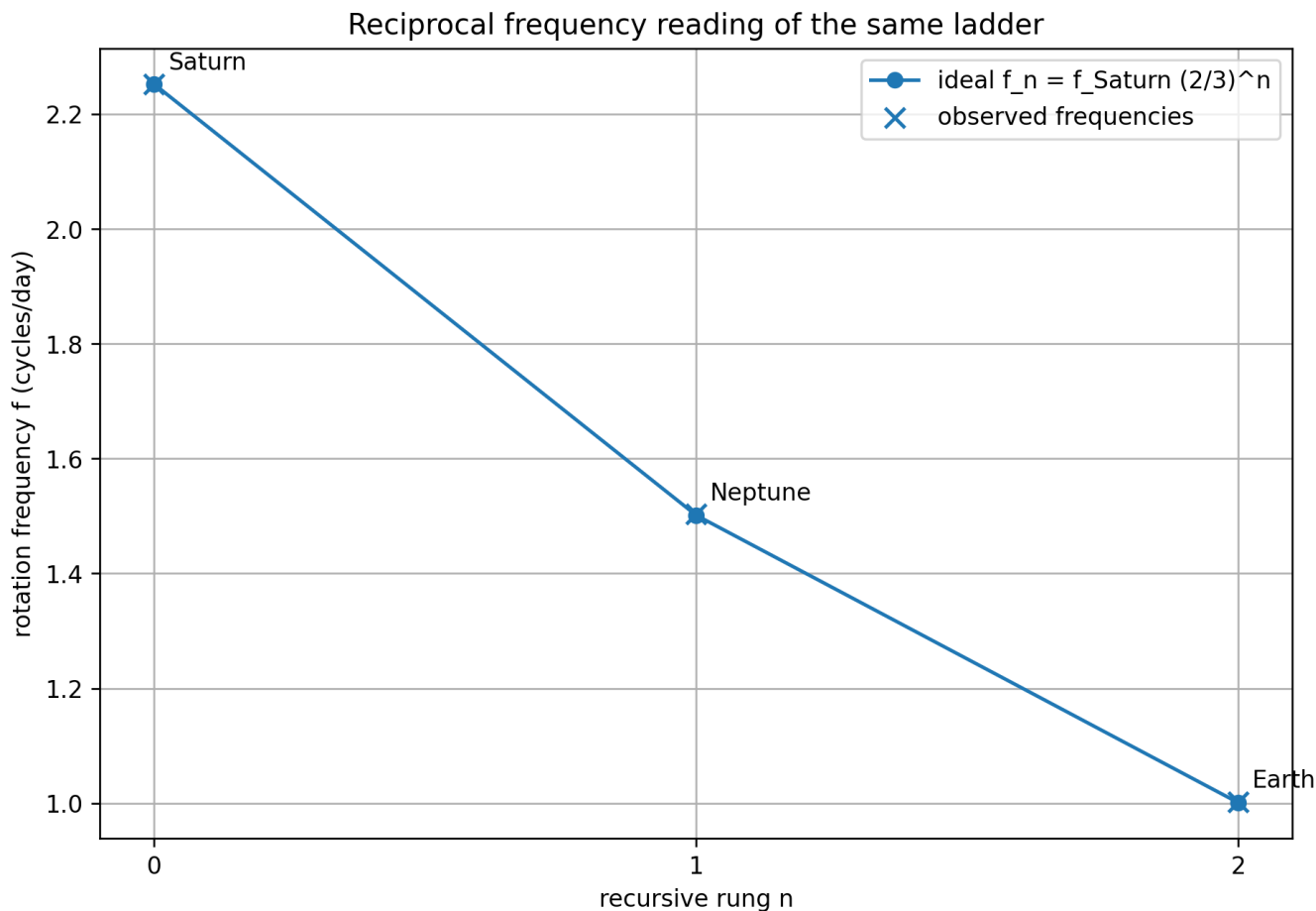


Figure 8. The same three bodies represented as cycles per day. The ideal curve is $f_n = f_{\text{Saturn}}(2/3)^n$.

Body	f (cycles/day)	ω (rad/day)	Ideal f	f residual (%)
Saturn	2.252205284444	14.151023151974	2.252205284444	+0.000000000
Neptune	1.503166043611	9.444670799469	1.501470189630	+0.112946231
Earth	1.002737843056	6.300387682440	1.000980126420	+0.175599554

9.2 Direct-reciprocal identity

$$(T_{(n+1)}/T_n) (f_{(n+1)}/f_n) = (3/2)(2/3) = 1.$$

The product equals unity because period and frequency are reciprocal coordinates. The ladder therefore preserves one invariant state transition viewed through two conjugate measures.

10. Extended rung occupancy and displaced states

10.1 Nearest assigned 3/2 rungs

After anchoring the ladder at Saturn, other bodies can be expressed by the logarithmic address $n(T)$. Earth and Neptune occupy the consecutive core states. Mars lies near the same $n=2$ rung as Earth. The adopted solar prime-meridian period lies near $n=10$. Mercury lies near $n=12$, although its physical branch is the independent 3:2 spin-orbit resonance.

The residual chart below is a direct reading of the assigned ladder; it does not redefine the branch membership.

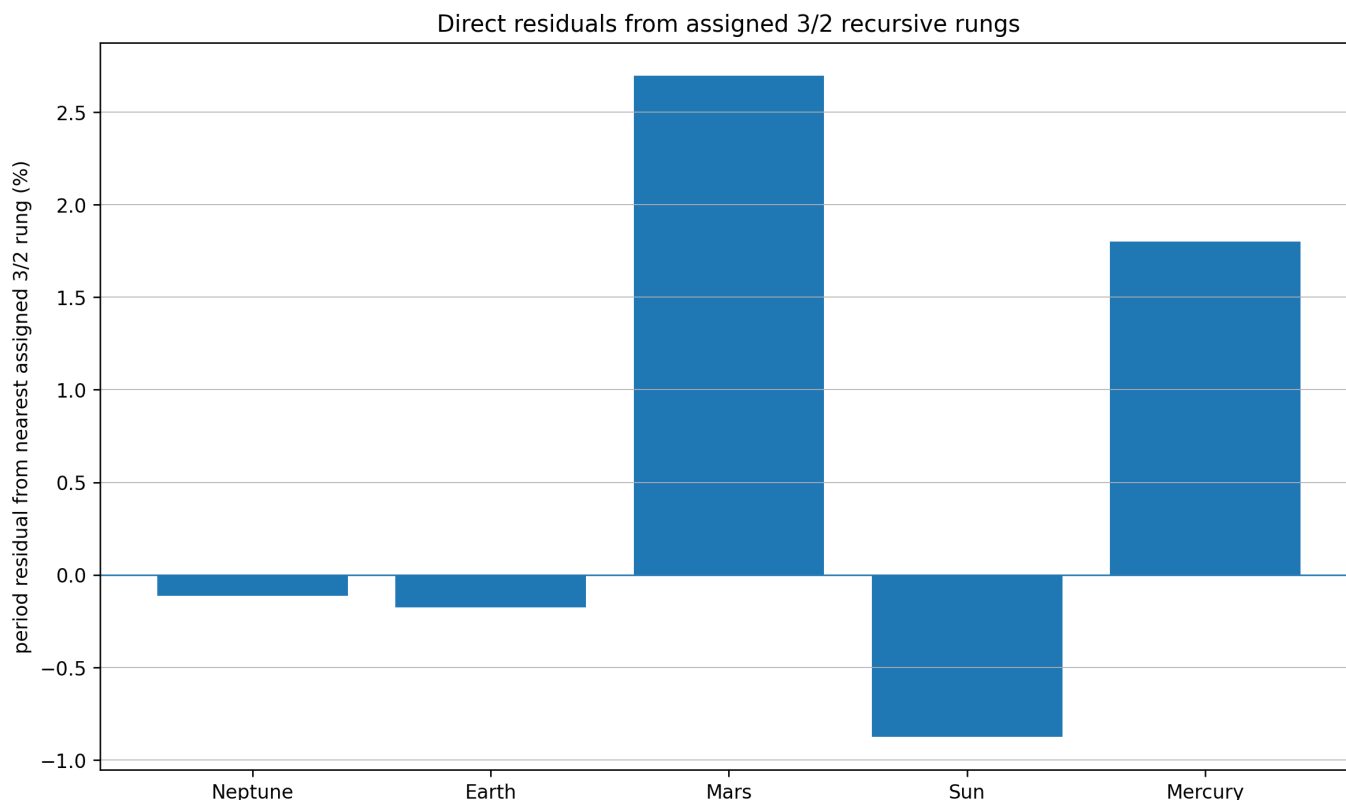


Figure 9. Signed period residual from each indicated nearest 3/2 rung. Neptune and Earth form the closest consecutive free-rotation sequence; Mars, the Sun, and Mercury are secondary rung neighbors.

Body	Assigned n	Actual T (day)	Ideal T (day)	Residual (%)	Residual (min)	Branch role
Saturn	0	0.444009259239	0.444009259239	+0.000000000	+0.0000	core free ladder
Neptune	1	0.665262499942	0.666013888858	-0.112818806	-1.0820	core free ladder
Earth	2	0.997269632263	0.999020833287	-0.175291742	-2.5217	core free ladder
Mars	2	1.025956755960	0.999020833287	+2.696232328	+38.7877	secondary n=2 neighbor
Sun	10	25.379994924001	25.603811278120	-0.874152491	-322.2955	secondary n=10 neighbor
Mercury	12	58.646145902358	57.608575375770	+1.801069580	+1494.1016	3:2 spin-orbit branch

10.2 Alternate-family requirement

Haumea, Ceres, Jupiter, Uranus, and Venus do not occupy the core Saturn-anchored ladder at the same residual scale. Their periods therefore require either another temporal anchor, a different recursive family, or a phase displacement coupled to body orientation and local field trajectory. This is the precise next classification problem.

11. Cell3 harmonic and MOD address transform

11.1 Coarse radial address

The period packet also applies the thirteen fixed Cell3 reciprocal radial amplitudes. Haumea is used as the reference period. For each body, a radial k , integer p , and denominator d in 1 through 16 define a coarse address:

$$\begin{aligned} R_i &= T_i/T_{\text{Haumea}} \\ R_{\text{hat}_i} &= (p_i/d_i) s_{\text{--}}(k_i) \\ u_i &= d_i R_i/s_{\text{--}}(k_i) \\ \text{delta}_i &= u_i - p_i. \end{aligned}$$

Nine of the 31 periods enter the original ± 10 ppm coarse envelope. The complete ledger is retained because the purpose here is to read address structure, not to force every body onto one common radial.

11.2 Fine MOD-81 coordinate

$$\text{delta}_i = q_i/81 + \rho_i.$$

The occupied classes are $q=-2,-1,0,+1,+2,+3$, with counts 1,4,20,4,1,1. The RMS within-class remainder is 0.003296814 and the maximum absolute remainder is 0.006095506.

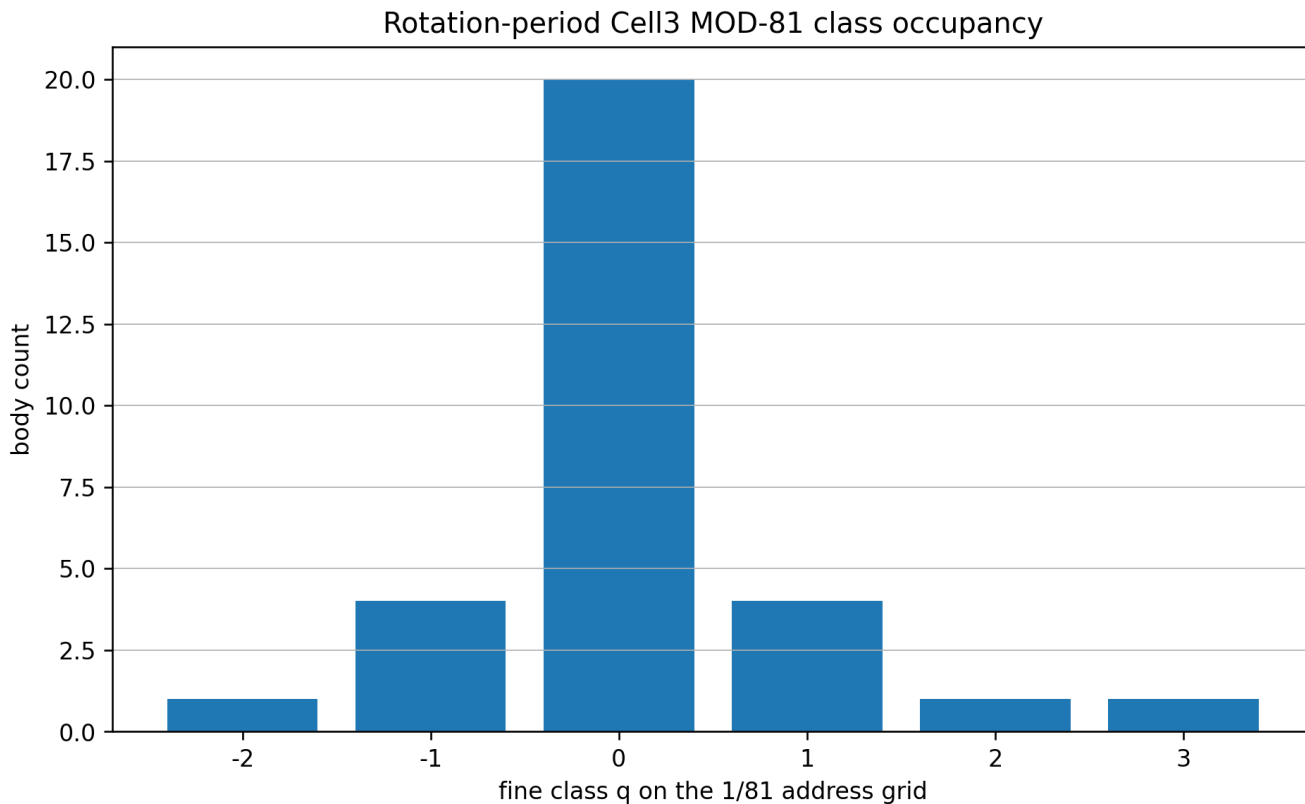


Figure 10. Observed MOD-81 class occupancy for all 31 rotation-period addresses. The central $q=0$ class contains 20 bodies.

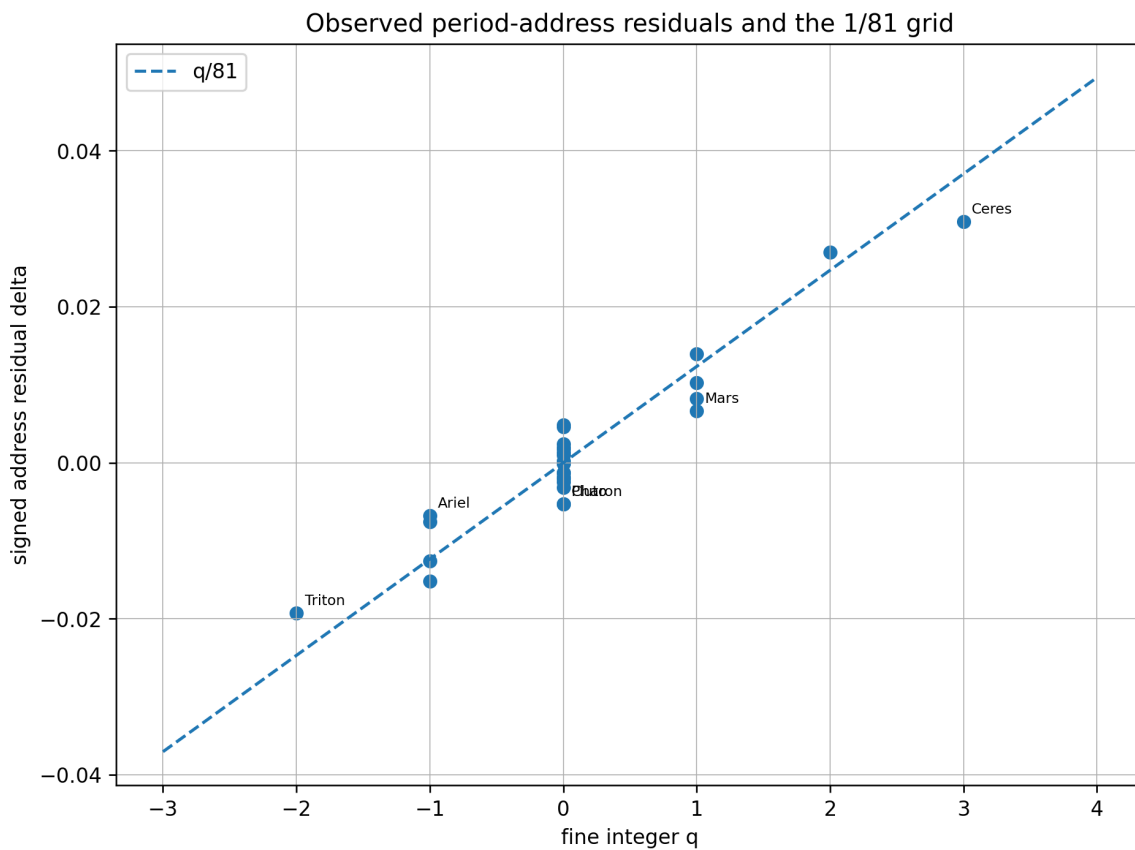


Figure 11. Signed coarse address residual delta against the nearest $q/81$ class. Labels mark the largest within-class remainders.

11.3 Interpretation of the MOD result

The period MOD spectrum is centrally concentrated rather than broadly distributed. This is consistent with the dataset architecture: twenty bodies are already constrained to synchronous 1:1 states, and the direct dynamical branches dominate the classification. The clearest temporal quantization is therefore the branch structure and the $3/2$ recursive ladder, while the MOD-81 layer remains a secondary address coordinate.

12. Geometric and physical interpretation

12.1 Why $3/2$ is native to the recursion

The ratio $3/2$ is not imported from an arbitrary musical or numerical sequence. It is generated by comparing the two subdivision scales already present in the Cell3 recursive grammar. A ternary interval is one third of its parent. A binary interval is one half. Their quotient is exactly $3/2$.

12.2 Temporal scale states

A body assigned to rung n carries a temporal address relative to the anchor T_0 . Moving one rung upward multiplies the period by $3/2$ and multiplies the frequency by $2/3$. Moving one rung downward multiplies the period by $2/3$ and the frequency by $3/2$. The address n is dimensionless and remains unchanged whether the period is measured in seconds, hours, or days.

12.3 Branch separation

Synchronous satellites occupy an orbitally imposed 1:1 branch. Mercury occupies an orbitally imposed 3:2 branch. Saturn, Neptune, and Earth occupy the free-rotation $3/2$ ladder. Keeping these branches distinct prevents tidal locking from being mistaken for an unconstrained period ladder while preserving the fact that all three branches are discrete rotational states.

12.4 K-Field reading

Within the K-Field framework, the Saturn-Neptune-Earth sequence is the temporal projection of the $3 \times 2 \times 2$ recursive scale relation. The period coordinate is the direct reading; the frequency coordinate is the reciprocal reading. The residual from the ideal rung is a phase displacement that can be compared with each body spin-axis orientation, orbital phase, and trajectory through the K-Field.

12.5 Conserved distinction between observation and mechanism

Observed: the periods and ratios reproduced in the ledger. Geometrically derived: the exact $3/2$ and $2/3$ factors from the subdivision grammar. Inferred structure: consecutive temporal rungs occupied by Saturn, Neptune, and Earth. Mechanistic program: relate the residual phase to orientation-dependent K-Field coupling and long-term rotational evolution.

13. Worked examples

13.1 Predicting Neptune from Saturn

$$\begin{aligned} T_{\text{Neptune, pred}} &= (3/2) T_{\text{Saturn}} \\ &= 1.5 \times 0.444009259239 \text{ day} \\ &= 0.666013888858 \text{ day.} \end{aligned}$$

The adopted Neptune period is 0.665262499942 day. The signed difference is -0.000751388917 day, or -1.082000 minutes.

13.2 Predicting Earth from Neptune

$$\begin{aligned} T_{\text{Earth, pred}} &= (3/2) T_{\text{Neptune}} \\ &= 1.5 \times 0.665262499942 \text{ day} \\ &= 0.997893749912 \text{ day.} \end{aligned}$$

The adopted Earth period is 0.997269632263 day. The signed difference is -0.000624117650 day, or -0.898729 minutes.

13.3 Predicting Earth directly from Saturn

$$\begin{aligned} T_{\text{Earth, pred}} &= (9/4) T_{\text{Saturn}} \\ &= 2.25 \times 0.444009259239 \text{ day} \\ &= 0.999020833287 \text{ day.} \end{aligned}$$

The adopted Earth period differs by -0.001751201025 day, or -2.521729 minutes.

13.4 Reciprocal frequency calculation

$$\begin{aligned} f_{\text{Saturn}} &= 1/T_{\text{Saturn}} = 2.252205284444 \text{ cycles/day} \\ (2/3) f_{\text{Saturn}} &= 1.501470189630 \text{ cycles/day} \\ f_{\text{Neptune}} &= 1.503166043611 \text{ cycles/day.} \end{aligned}$$

The same calculation repeated from Neptune predicts the Earth frequency. The direct and reciprocal residuals have opposite sign because frequency inverts period.

14. Discovery program opened by the temporal ladder

14.1 Residual phase versus spin-axis orientation

For each non-synchronous body define the logarithmic rung phase $r_T = n(T) - \text{round}(n(T))$. Compare r_T with the body spin-axis position expressed in Cell3 coordinates. A systematic relation would identify whether temporal displacement is controlled by plane-family phase, phase-cost gradient, or orientation relative to the CMB traversal vector.

$$r_T = \log(T/T_0)/\log(3/2) - \text{round}[\log(T/T_0)/\log(3/2)].$$

14.2 Secular drift prediction

Long-baseline rotation measurements can test whether bodies move toward, away from, or around the recursive rung centers. The sign of dT/dt supplies a time-odd observable. Coupled with the signed axis orientation, it can distinguish two otherwise antipodal spin branches.

14.3 Alternate temporal families

Bodies not on the Saturn-anchored sequence should be searched against the thirteen reciprocal Cell3 radial amplitudes and against anchors defined by the other free rotators. The objective is not unrestricted numerical fitting. The objective is to identify a small set of geometrically generated temporal families with explicit branch rules.

14.4 Satellite systems as controlled laboratories

Synchronous satellites provide known 1:1 boundary conditions, while resonant satellite pairs provide known 2:1 and 4:2:1 transitions. These systems can be used to calibrate how orbital forcing, tidal dissipation, and K-Field temporal addressing combine without conflating the mechanisms.

14.5 Solar-system extension

The same procedure extends to additional moons, asteroids, dwarf planets, and measured differential-rotation components. Every added body must carry a declared period convention, spin sign, branch type, source epoch, and uncertainty so that the temporal address remains physically interpretable.

15. Determination

The rotation-period structure is organized first by discrete dynamical branches. Twenty bodies occupy synchronous or mutually synchronous 1:1 states. Mercury occupies the 3:2 spin-orbit state. The Galilean satellites inherit a 1:2:4 rotational chain from their synchronous orbital resonance. The Saturnian pairs reproduce 2:1 structure. Pluto and Charon reproduce the mutual 1:1 state.

Inside the non-synchronous free-rotation branch, Saturn, Neptune, and Earth occupy consecutive temporal rungs generated by $T_{(n+1)} = (3/2)T_n$. The first measured ratio is 1.498307718; the second is 1.499061848; and Earth/Saturn is 2.246055936, the two-rung 9/4 state. In frequency space the same sequence descends by 2/3 per rung.

The factor 3/2 is the direct quotient of the binary and ternary interval scales in the Cell3 3 x 2 x 2 recursive subdivision. The 2/3 frequency factor is its reciprocal. The Saturn-Neptune-Earth sequence is therefore the temporal form of the recursive geometry: period carries the direct scale address and frequency carries the reciprocal scale address.

The principal next measurement is the signed relationship between temporal rung residual, spin-axis Cell3 phase, and secular rotational drift. That comparison converts the discovered ladder from a static period sequence into a dynamical probe of K-Field coupling.

Appendix A. Complete solar-system rotation-period ledger

Body	W_dot (deg/day)	T (day)	T (hour)	Branch	Cell3 radial	p/d	q	rho
Haumea	+2207.2348252606	0.163100000000	3.914400000	free rotation	001	1/1	0	+0.000000000
Ceres	+952.1532000000	0.378090416542	9.074169997	free rotation	010	19/11	3	-0.006095506
Jupiter	+870.5360000000	0.413538325813	9.924919820	free rotation	011	29/12	1	-0.002075997
Saturn	+810.7939024000	0.444009259239	10.656222222	free rotation	01-1	9/7	-1	+0.004799175
Neptune	+541.1397757000	0.665262499942	15.966299999	free rotation	011	35/9	0	+0.001794262
Uranus	-501.1600928000	0.718333333344	17.240000000	free rotation	110	31/16	0	-0.001986632
Mimas	+381.9945550000	0.942421810175	22.618123444	synchronous 1:1	010	69/16	0	-0.001884178
Earth	+360.9856235000	0.997269632263	23.934471174	free rotation	1-1-1	11/4	0	+0.001032314
Mars	+350.8919824433	1.025956755960	24.622962143	free rotation	10-1	20/7	1	-0.005681093
Enceladus	+262.7318996000	1.370218083712	32.885234009	synchronous 1:1	1-11	34/11	0	-0.002445404
Miranda	-254.6906892000	1.413479232911	33.923501590	synchronous 1:1	10-1	63/16	0	+0.002387904
Io	+203.4889538000	1.769137799754	42.459307194	synchronous 1:1	10-1	69/14	0	-0.001874378
Tethys	+190.6979085000	1.887802560771	45.307261459	synchronous 1:1	01-1	71/13	0	+0.004856963
Ariel	-142.8356681000	2.520378871669	60.489092920	synchronous 1:1	010	173/15	-1	+0.005536692
Dione	+131.5349316000	2.736915552553	65.685973261	synchronous 1:1	011	16/1	0	-0.000141115
Europa	+101.3747235000	3.551181079177	85.228345900	synchronous 1:1	010	65/4	0	-0.001291152
Umbriel	-86.8688923000	4.144176246161	99.460229908	synchronous 1:1	1-1-1	80/7	0	+0.001312401
Rhea	+79.6900478000	4.517502623458	108.420062963	synchronous 1:1	1-10	195/16	-1	-0.002868696
Triton	-61.2572637000	5.876854078286	141.044497879	synchronous 1:1	011	481/14	-2	+0.005444275
Charon	+56.3625225000	6.387222999113	153.293351979	synchronous 1:1	1-10	224/13	0	-0.005290653
Pluto	+56.3625225000	6.387222999113	153.293351979	synchronous 1:1	1-10	224/13	0	-0.005290653
Ganymede	+50.3176081000	7.154553119547	171.709274869	synchronous 1:1	01-1	207/10	0	+0.000143343
Titania	-41.3514316000	8.705865457872	208.940770989	synchronous 1:1	01-1	403/16	1	+0.001593577
Oberon	-26.7394932000	13.463231980777	323.117567539	synchronous 1:1	001	908/11	0	+0.004609372
Titan	+22.5769768000	15.945447576489	382.690741836	synchronous 1:1	011	1305/14	2	+0.002331395
Callisto	+21.5710715000	16.689017974837	400.536431396	synchronous 1:1	01-1	338/7	0	+0.000199135
Sun	+14.1844000000	25.379994924001	609.119878176	free rotation	1-10	1027/15	-1	-0.000245176
Moon	+13.1763581500	27.321661714242	655.719881142	synchronous 1:1	100	1186/13	1	-0.004091086
Mercury	+6.1385108000	58.646145902358	1407.507501657	Mercury 3:2	001	2517/7	0	+0.001970058
Iapetus	+4.5379572000	79.330849572579	1903.940389742	synchronous 1:1	10-1	221/1	0	-0.001309271
Venus	-1.4813688000	243.018483985892	5832.443615661	free rotation	001	1490/1	0	-0.003163790

Appendix B. Complete 27-node Cell3 coordinate ledger

Address	Category	ICRS x (r)	ICRS y (r)	ICRS z (r)
(-1,-1,-1)	corner	-0.010974902499961	+0.007385983127423	+0.039661911011939
(-1,-1,0)	edge center	-0.011290092242635	-0.004742877881141	+0.018199733726392
(-1,-1,1)	corner	-0.011605281985309	-0.016871738889705	-0.003262443559155
(-1,0,-1)	edge center	-0.009875529515800	+0.013837068256769	+0.022628660243911
(-1,0,0)	face center	-0.010190719258473	+0.001708207248205	+0.001166482958364
(-1,0,1)	edge center	-0.010505909001147	-0.010420653760359	-0.020295694327182
(-1,1,-1)	corner	-0.008776156531638	+0.020288153386115	+0.005595409475884
(-1,1,0)	edge center	-0.009091346274312	+0.008159292377552	-0.015866767809663
(-1,1,1)	corner	-0.009406536016985	-0.003969568631012	-0.037328945095210
(0,-1,-1)	edge center	-0.000784183241488	+0.005677775879218	+0.038495428053575
(0,-1,0)	face center	-0.001099372984162	-0.006451085129346	+0.017033250768028
(0,-1,1)	edge center	-0.001414562726835	-0.018579946137910	-0.004428926517519
(0,0,-1)	face center	+0.000315189742674	+0.012128861008564	+0.021462177285547
(0,0,0)	body center	+0.000000000000000	+0.000000000000000	+0.000000000000000
(0,0,1)	face center	-0.000315189742674	-0.012128861008564	-0.021462177285547
(0,1,-1)	edge center	+0.001414562726835	+0.018579946137910	+0.004428926517519
(0,1,0)	face center	+0.001099372984162	+0.006451085129346	-0.017033250768028
(0,1,1)	edge center	+0.000784183241488	-0.005677775879218	-0.038495428053575
(1,-1,-1)	corner	+0.009406536016985	+0.003969568631012	+0.037328945095210
(1,-1,0)	edge center	+0.009091346274312	-0.008159292377552	+0.015866767809663
(1,-1,1)	corner	+0.008776156531638	-0.020288153386115	-0.005595409475884
(1,0,-1)	edge center	+0.010505909001147	+0.010420653760359	+0.020295694327182
(1,0,0)	face center	+0.010190719258473	-0.001708207248205	-0.001166482958364
(1,0,1)	edge center	+0.009875529515800	-0.013837068256769	-0.022628660243911
(1,1,-1)	corner	+0.011605281985309	+0.016871738889705	+0.003262443559155
(1,1,0)	edge center	+0.011290092242635	+0.004742877881141	-0.018199733726392
(1,1,1)	corner	+0.010974902499961	-0.007385983127423	-0.039661911011939

Appendix C. Plane and recursive-boundary ledger

Object	Equation or coordinate	Value
n_a	ICRS row 1	(+0.995965190308961, -0.083602162507377, +0.032619290564400)
delta_a	native spacing at r=1	0.020508723238107814
a- boundary	n_a dot x	-0.010254361619053907
a+ boundary	n_a dot x	+0.010254361619053907
n_b	ICRS row 2	(+0.089052594101605, +0.866576769291032, -0.491033948326276)
delta_b	native spacing at r=1	0.028104333806782844
b- boundary	n_b dot x	-0.014052166903391422
b+ boundary	n_b dot x	+0.014052166903391422
n_c	ICRS row 3	(-0.194092200807768, -0.913177168821229, -0.358379234790804)
delta_c	native spacing at r=1	0.037657147000386110
c- boundary	n_c dot x	-0.018828573500193055
c+ boundary	n_c dot x	+0.018828573500193055
a-subdivision boundaries	address coordinate a	-1/2, -1/6, +1/6, +1/2
b-subdivision boundaries	address coordinate b	-1/2, 0, +1/2
c-subdivision boundaries	address coordinate c	-1/2, 0, +1/2
subcell count	3 x 2 x 2	12
direct ratio	(1/2)/(1/3)	3/2
reciprocal ratio	(1/3)/(1/2)	2/3

Appendix D. Machine reconstruction contract

The companion JSON file is the authoritative machine-readable packet for this edition. It contains the complete source hashes, Cell3 basis, plane normals, spacings, 27-node ledger, 31-body period ledger, resonance ratios, 3/2 ladder assignments, MOD-81 classes, equations, generated-figure hashes, and report metadata.

Minimal reconstruction order

1. Read B, N, and delta from the machine packet without reordering labels.
2. Verify $\det(B) < 0$ and $N B = \text{diag}(\delta_a, \delta_b, \delta_c)$.
3. Generate the centered nodes $p = B[i, j, k]^T / 2$ for i, j, k in $\{-1, 0, 1\}$.
4. Generate six face planes $n_i \cdot x = \pm \delta_i / 2$.
5. Generate the $3 \times 2 \times 2$ address boundaries $a = \{-1/2, -1/6, +1/6, +1/2\}$, $b = c = \{-1/2, 0, +1/2\}$.
6. Compute periods from $T = 360 / |W_{\text{dot}}|$, except Haumea, whose direct period is supplied.
7. Set $T_0 = T_{\text{Saturn}}$ and compute $n = \log(T / T_0) / \log(3/2)$.
8. Reconstruct ideal period and frequency ladders.
9. Read the direct resonance ratios and MOD-81 address fields.
10. Preserve all native units and apply physical rendering only at the terminal stage.

Machine-readable file

K-Field_3_2_2_Solar_system_bodies_rotation_period_harmonic_ladder_1784893330.json

Report file

K-Field_3_2_2_Solar_system_bodies_rotation_period_harmonic_ladder_1784893330.pdf

References and source ledger

ID	Source
K1	cell3_natural_unit_ICRS_embedding_v1.0.json. Canonical native Cell3 ICRS embedding, basis, plane normals, spacings, node rule, and validation contract.
K2	K_Field_solar_system_rotation_period_quantization_MOD_1784839005.json. Complete 31-body period, harmonic, ladder, and MOD packet used for this report.
K3	K-Field_mass_harmonic_mod_quantization_1784580520.pdf. Historical archival format and Cell3 radial/MOD grammar reference.
K4	solar_system_body_axis_orientations_ICRS_J2000(1).txt. Body-set and positive-angular-velocity pole convention.
1	NASA Navigation and Ancillary Information Facility. Generic planetary constants kernel pck00011.tpc. https://naif.jpl.nasa.gov/pub/naif/generic_kernels/pck/pck00011.tpc
2	JPL Solar System Dynamics. Planetary Physical Parameters. https://ssd.jpl.nasa.gov/planets/phys_par.html
3	NASA Science. Mercury: Facts. https://science.nasa.gov/mercury/facts/
4	NASA Science. Ganymede: Facts. https://science.nasa.gov/jupiter/jupiter-moons/ganymede/facts/
5	NASA Science. Enceladus. https://science.nasa.gov/saturn/moons/enceladus/
6	NASA Science. Charon. https://science.nasa.gov/dwarf-planets/pluto/moons/charon/
7	NASA Science. Tidal Locking. https://science.nasa.gov/moon/tidal-locking/

End of archival computational edition.